Theorem:-

Let $T : U \to V$, The set $L(U,V)$ of all Linear transformations from U to V together with the operations of addition and scalarr multiplication is a vectorspace.

Proof:-

Given $L(U,V)$: The set of all Linear transformations from U to V.

(i) Let $T_1, T_2 \in L(U,V)$ such that

$T_1(u_1) = v_1$

$T_2(u_2) = v_2$

$$(T_1 + T_2)(u) = T_1(u) + T_2(u)$$
$$= v_1 + v_2 \in V$$

$\therefore T_1 + T_2$ is a Linear transformation.

(ii) Let $T_1, T_2, T_3 \in L(U,V)$ such that

$T_1(u) = v_1$

$T_2(u) = v_2$

$T_3(u) = v_3$

$$\{(T_1 + T_2) + T_3\}(u) = (T_1 + T_2)(u) + T_3(u)$$
$$= T_1(u) + T_2(u) + T_3(u)$$
$$= T_1(u) + (T_2 + T_3)(u)$$
$$= T_1 + (T_2 + T_3)$$

(iii) $\exists$ a zero map $0 : U \to V$ such that

$$(T_1 + 0)(u) = T_1(u) + 0(u)$$
$$= v_1 + 0$$
$$= v_1$$
$$= T_1(u)$$

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